

# **K.A.L.A KCSE 2025 PREDICTION**

***(KENYA HIGH-ALLIANCE -LENANA-ALLIANCE GIRLS)***

## **CHEMISTRY**

**233/3**

**PAPER 3 (PRACTICAL)**

**TIME: 2¼ HOURS**

**NAME.....**

**SCHOOL..... SIGN.....**

**INDEX NO..... ADM NO.....**

### ***Kenya Certificate of Secondary Education.***

#### **Instructions to Candidates:**

- 1. Write your name and index number in the spaces provided.*
- 2. Sign and write the date of examination in the spaces provided above.*
- 3. Answer **ALL** questions in the spaces provided.*
- 4. You are required to spend the first 15 minutes of the 2 ¼ hours allowed for this  
Paper reading the whole paper carefully before commencing your work.*
- 5. Additional pages must not be inserted.*

#### **FOR THE EXAMINER'S USE ONLY**

| <i>Question</i>    | <i>Maximum score</i> | <i>Candidate's Score</i> |
|--------------------|----------------------|--------------------------|
| 1                  | 21                   |                          |
| 2                  | 13                   |                          |
| 3                  | 06                   |                          |
| <b>TOTAL SCORE</b> | <b>40</b>            |                          |

1. You are provided with;
  - 3.6 g of solid P in a boiling tube. Solid P is a hydrated dibasic acid with the formula  $\text{H}_2\text{C}_2\text{O}_4 \cdot n\text{H}_2\text{O}$
  - Solution L which is a 0.2 M sodium hydroxide solution. You are required to determine;
    - (i) Solubility of solid P
    - (ii) The value of n in the formula  $\text{H}_2\text{C}_2\text{O}_4 \cdot n\text{H}_2\text{O}$

### Procedure I

- I. Using a burette, add  $4\text{cm}^3$  of distilled water to solid P in the boiling tube. Heat the mixture while stirring with the thermometer to about  $70^\circ\text{C}$ . When all of solid has dissolved, allow the solution to cool while stirring with the thermometer. Note the temperature at which crystals of solid P first appear. Record the temperature in **table 1** below.
- II. Using the burette, add  $2\text{cm}^3$  of distilled water to the contents of the boiling tube. Warm the mixture while stirring with the thermometer until all the solid dissolves again. Allow the mixture to cool while stirring. Note the temperature at which crystals of solid P appears.
- III. Repeat procedure (II) three more times and record the temperatures in **table 1**. **RETAIN the contents of the boiling tube for use in PROCEDURE II**

TABLE 1

| Volume of water in the boiling tube ( $\text{cm}^3$ ) | Crystallization temperature ( $^\circ\text{C}$ ) | Solubility of solid P in g/100 g of water |
|-------------------------------------------------------|--------------------------------------------------|-------------------------------------------|
| 4                                                     |                                                  |                                           |
| 6                                                     |                                                  |                                           |
| 8                                                     |                                                  |                                           |
| 10                                                    |                                                  |                                           |
| 12                                                    |                                                  |                                           |

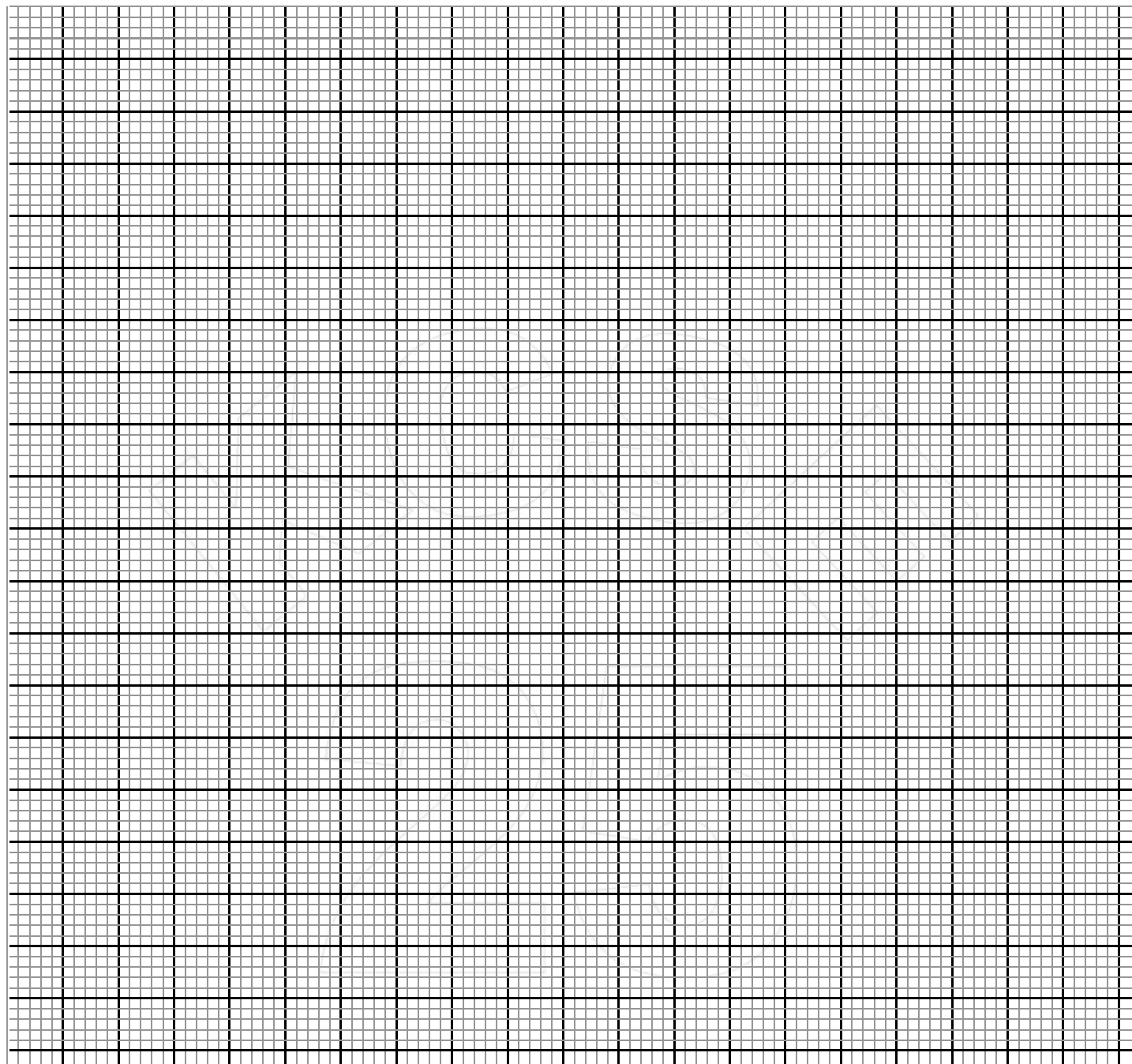
(4marks)

- (a) Complete **table 1** above by working out the solubility.

(1 mark)



On the graph provided, plot a graph of solubility of solid P against crystallization temperature. (3mk)



(b) From the graph, determine;

(i) The solubility of solid P at 50 °C

(1 mark)

.....

(ii) The temperature at which 65 g of solid P would dissolve in 100 g of water (1 mark)

.....

.....



KCSE  
25

## **Procedure II**

- (i) Transfer the contents of the boiling tube from **PROCEDURE I** into a clean 250 ml volumetric flask.
- (ii) Add distilled water up to the mark
- (iii) Label the resulting solution as solution P
- (iv) Fill the burette with solution P
- (v) Pipette  $25\text{cm}^3$  of solution L into a conical flask. Add three drops of phenolphthalein indicator
- (vi) Titrate solution P against solution L to an accurate end point. Record your results in **table 2** below.

| <b>TABLE 2</b>                              | 1 | 2 | 3 |
|---------------------------------------------|---|---|---|
| Final burette reading                       |   |   |   |
| Initial burette reading                     |   |   |   |
| Volume of solution P used ( $\text{cm}^3$ ) |   |   |   |

**(4 marks)**

- a) Calculate Average volume of solution P used.

**(1 mark)**

- b) (i) Moles of solution L used.

**(1 mark)**

- (ii) Moles of solution P used.

**(1 mark)**

- (iii) Moles of solution P in  $250\text{ cm}^3$  of solution P

**(1 mark)**

- (iv) The relative formula mass of P.

**(1 mark)**

(c) Determine the value of n in the formula;  $\text{H}_2\text{C}_2\text{O}_4 \cdot n\text{H}_2\text{O}$  (H = 1, C = 12, O = 16)

(2marks)

2. You are provided with solid E. Carry out the experiments below. Write your observations and inferences in the spaces provided.

Place all solid E into a boiling tube. Add about  $20\text{cm}^3$  of distilled water and shake. Retain the contents of the boiling tube.

| <i>Observations</i> | <i>Inferences</i> |
|---------------------|-------------------|
|                     |                   |

Use  $2\text{cm}^3$  of solution E, in a test tube in each experiment i, ii,iii,iv and v

i. To experiment i, Add two drops of aqueous Sulphuric vi acid

| <i>Observations</i> | <i>Inferences</i> |
|---------------------|-------------------|
|                     |                   |

(1 mark)

(1 mark)

ii. To experiment ii, add NaOH solution drop wise till in excess.



3). You are provided with liquid Q. carry out the tests below. Write your observations and inferences in the spaces provided.

i). To 2 cm<sup>3</sup> of liquid Q in a test tube, add universal indicator

| <i>Observations</i> | <i>Inferences</i> |
|---------------------|-------------------|
|                     |                   |

(1 mark)

(1 mark)

ii). Place 3 drops of liquid Q on a watch glass and ignite.

| <i>Observations</i> | <i>Inferences</i> |
|---------------------|-------------------|
|                     |                   |

(1 mark)

(1 mark)

iii). To 2 cm<sup>3</sup> of liquid Q in a test tube , add two or three drops of acidified potassium dichromate VI and warm

| <i>Observations</i> | <i>Inferences</i> |
|---------------------|-------------------|
|                     |                   |

(1 mark)

(1 mark)